C4.10 Handling overlapping write transactions

This section describes the expected behavior when two masters attempt stores to the same cache line in a shareable region of memory at approximately the same time. When this happens, it is the responsibility of the interconnect to sequence the order in which the transactions occur.

The master that gets sequenced first proceeds with the transaction as normal, however the master that is sequenced second sees the transactions associated with the first master's store on its snoop port while attempting to carry out a store.

The following sections describe the expected behavior from the standpoint of the master that is sequenced second. For brevity, the master that is sequenced first is referred to as master1 and the master that is sequenced second is referred to as master2.

C4.10.1 Overlapping ReadUnique

If master2 has issued a ReadUnique transaction because it required a copy of the data, the following occurs:

1. Master2 issues a ReadUnique transaction.
2. Master2 then sees one of the following transactions on its snoop port from master1 attempting a write to the same line:
   - ReadUnique
   - CleanInvalid
   - MakeInvalid.

   At this point master2 must invalidate any local copy it has of the cache line. If master2 does not have a local copy of the cache line then no action is required.
3. When the ReadUnique completes, it returns with the updated copy of the cache line that includes the store performed by master1.
4. Master2 can perform its store.

C4.10.2 Overlapping MakeUnique

If master2 has issued a MakeUnique transaction because it was performing a full cache line write, the following occurs:

1. Master2 issues a MakeUnique transaction.
2. Master2 sees one of the following transactions on its snoop port from master1 attempting a write to the same line:
   - ReadUnique
   - CleanInvalid
   - MakeInvalid.

   At this point master2 must invalidate any local copy it has of the cache line. If master2 does not have a local copy of the cache line then no action is required.
3. When the MakeUnique completes, Master2 can perform its full cache line store.

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C4.10.3 Overlapping CleanUnique

If master2 has issued a CleanUnique transaction because it was performing a partial line store but it already had a cached copy of the line, the following occurs:

1. Master2 issues a CleanUnique transaction.
2. Master2 sees one of the following transactions on its snoop port from master1 attempting a write to the same line:
   - ReadUnique
   - CleanInvalid
   - MakeInvalid.

   At this point master2 must respond to the snoop appropriately and then invalidate its local copy of the cache line.
3. When the CleanUnique completes, Master2 cannot perform its local store because it has lost its local copy of the cache line.
4. Master2 can issue a new ReadUnique transaction to obtain a copy of the line.
5. Master2 can perform its store.

A master can remove the need to issue a new ReadUnique the transaction, as described in the CleanUnique case, by initially issuing a ReadUnique transaction instead of a CleanUnique transaction. However, this sometimes results in a fetch from main memory occurring when it is not required.

Alternatively, a master can remove the need to issue a new ReadUnique transaction by performing a partial line WriteBack to main memory, that only updates the required bytes, when its CleanUnique transaction completes and the master has permission to store to the line. This does mean that the master does not retain a copy of the line.

It is acceptable for a master to use a CleanUnique transaction when carrying out a full cache line store. In this case, the master does not have to retry the transaction with a ReadUnique, it can simply perform the full cache line store when the CleanUnique is complete.

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Chapter C5
Snoop Transactions

This chapter describes the snoop transactions seen on the snoop address channel. Both the required and protocol-recommended snoop transaction behaviors are described. It contains the following sections:

- *Mapping coherency operations to snoop operations* on page C5-210
- *General requirements for snoop transactions* on page C5-213
- *Snoop transactions* on page C5-219.

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